

IB MATH AA SL

Lesson 30

Discrete Random Variables

A large, stylized, light blue letter 'X' is positioned on the right side of the slide. It is set against a dark blue background that features two large, overlapping circles of a slightly lighter shade of blue. The 'X' is composed of two intersecting diagonal lines with thick strokes.

Statistics & Probability · 60-minute lesson

Syllabus SL 4.7 · Property of Lynda Badmus Education

Do Now — recall

ON YOUR WHITEBOARD

1. List the outcomes of one die roll.
2. What must all probabilities add to?
3. $P(X=2)$ if X is a fair die?

Answers (click to reveal)

1. 1,2,3,4,5,6
2. 1
3. $1/6$

What we're learning today

LEARNING OBJECTIVES

- 1 Understand**
a probability distribution.
- 2 Check**
that probabilities sum to 1.
- 3 Calculate**
the expected value $E(X)$.
- 4 Interpret**
 $E(X)$ as a long-run average.

SUCCESS CRITERIA — I CAN...

- ✓
read a probability distribution table
- ✓
use $\sum P(X = x) = 1$ to find a missing value
- ✓
compute $E(X) = \sum x \cdot P(X = x)$
- ✓
explain what $E(X)$ means

Distributions

PROBABILITIES SUM TO 1

$$\sum P(X = x) = 1$$

Distribution

lists each value with its probability

Missing value

use the sum = 1 to find it

EXAMPLE

x

1, 2, 3

P(X=x)

0.2, 0.5, ?

?

= 1 - 0.7 = 0.3

Find a missing probability

QUESTION

$x : 1, 2, 3, 4$

$P(X=x) : 0.1, 0.3, k, 0.2$

Find k .

SOLUTION

All probabilities add to 1:

$$0.1 + 0.3 + k + 0.2 = 1$$

So $0.6 + k = 1$:

$$\mathbf{k = 0.4}$$

Your turn — sum to 1

ANSWER IN YOUR BOOK

1. $P: 0.2, 0.2, k$. Find k .
2. $P: 0.15, k, 0.35, 0.2$. Find k .
3. Can a probability be 1.2?

Answers (click to reveal)

1. 0.6
2. 0.3
3. No (must be 0–1)

Expected value

THE MEAN OF X

$$E(X) = \sum x P(X=x)$$

E(X)

the long-run average value of X

Method

multiply each x by its probability, then add

MEANING

Not necessarily a possible value.

E.g. a die has $E(X) = 3.5$.

Use

fair-game and decision problems.

Expected value

QUESTION

$x : 0, 1, 2$

$P(X=x) : 0.5, 0.3, 0.2$

Find $E(X)$.

SOLUTION

$E(X) = \sum x \cdot P(X=x):$

$$E(X) = 0(0.5) + 1(0.3) + 2(0.2)$$

Evaluate:

$$E(X) = 0.7$$

Exam-style question

The random variable X has $P(X=x) = 0.1, 0.4, 0.3, 0.2$ for $x = 1, 2, 3, 4$.

- (a) Verify the probabilities sum to 1. [1]
- (b) Find $E(X)$. [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $0.1+0.4+0.3+0.2 = 1$ (A1)
- (b) $E(X) = 1(0.1)+2(0.4)+3(0.3)+4(0.2)$ (M1)(A1) $= 2.6$ (A1)

Common mistakes



Sum to 1

Every probability distribution must total 1.



$E(X)$ method

Multiply each value by ITS probability, then add.



$E(X)$ possible?

$E(X)$ need not be an attainable value.



Range

Each probability is between 0 and 1.

Mixed practice

QUESTIONS

1. $P: 0.25, 0.25, k$. Find k .
2. $E(X)$ for $x=1,2$ with $P=0.5,0.5$.
3. $E(X)$ for $x=0,10$ with $P=0.9,0.1$.
4. Sum of all probabilities = ?

Answers (click to reveal)

1. 0.5
2. 1.5
3. 1
4. 1

Plenary & exit ticket

KEY FORMULAS

$$\sum P(X=x)=1$$

$$E(X)=\sum x P(X=x)$$

EXIT TICKET — before you leave

1. P: 0.3, k, 0.5. Find k.
2. E(X): x=2,4; P=0.5,0.5.
3. Can P = -0.1?

Answers (click to reveal)

1. 0.2
2. 3
3. No

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Discrete random variables.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 31

The binomial distribution — repeated trials.

$$X \sim B(n, p)$$